

Misallocation and Transition Dynamics in the Restuccia–Rogerson Framework

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Abstract

This thesis develops and analyzes a dynamic, two-asset general equilibrium extension of the Restuccia and Rogerson (2008) misallocation framework. The original framework lets wedges distort the allocation of labor and capital across establishments within a stationary equilibrium. The model constructed here adopts those distortions into a forward-looking system with intertemporal optimization. Firms are treated as a second asset held by households, alongside physical capital.

This dynamic structure introduces new channels through which reforms propagate. Distortions affect not only plant-level allocations but also the intertemporal returns to equity and capital and the evolution of the firm mass. The resulting transitional dynamics are therefore shaped by such endogenous factors.

It is then shown that welfare depends on the entire transition path, which can produce welfare measurements different from the steady-state comparison. Results from this model’s implied transition path show that there is welfare gain from removing distortions, but the gain is less than when only comparing steady-states.

More complex versions of this framework could be useful for evaluating reform sequences, distinguishing between anticipated and unanticipated policy changes, and quantifying welfare effects. The extension thus offers a model of how misallocation interacts with aggregate dynamics.

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1 Introduction

A body of empirical research links cross-country income differences to the misallocation of resources across heterogeneous establishments. Restuccia and Rogerson (2008) formalized this idea in a plant-level model in which idiosyncratic wedges distort the allocation of capital and labor. Their central finding, that modest dispersion in establishment-level wedges can generate large reductions in aggregate productivity, is what motivates this analysis.

The original RR paper focuses on comparing steady states, which is sufficient to make their point. It describes how distortions map into a stationary allocation, but it does not describe how economies adjust when distortions change. I am interested in characterizing the transitions, as these may matter for welfare questions. Forward-looking households and firms anticipate future reforms. Because welfare depends on the entire consumption path a dynamic view of misallocation is useful for understanding the consequences of reform.

This thesis provides such a dynamic extension. I reconstruct the RR model, then embed it in a general equilibrium environment in which households choose both physical capital and equity claims to firms. Firms are treated as a second asset: they pay dividends reflecting operating profits and survive each period with probability $(1 - \lambda)$. The price of equity is determined endogenously through a no-arbitrage condition equating the current price to the discounted value of dividends and continuation values under household beliefs. Entry is determined by a forward-looking free-entry condition equating the cost of creating a firm to the expected discounted value of its equity.

The result is a dynamic model in which distortions affect not only static allocations but also the evolution of the productive sector over time. Because the transition path must satisfy both forward-looking household behavior and backward-recursive asset pricing, the equilibrium is pinned down by a system linking plant-level decisions, entry dynamics, equity valuation, and capital accumulation. This is a structure for studying reform and welfare in environments with establishment-level distortions.

This thesis briefly reviews my replication and then details the theory of the extension. I discuss the numerical implementation as well. The goal is to discuss some dynamic channels through which misallocation interacts with investment and firm turnover, thereby offering a consistent transition analysis.

2 Related Literature

Here I spend some time going over some other literature that motivates analysis of misallocation with firm dynamics.

An interdisciplinary literature shows that resource misallocation across heterogeneous producers is a first-order determinant of cross-country differences in productivity and income. Restuccia and Rogerson (2008) provided an articulation of this idea: even in a setting with standard technology, no aggregate price distortions, and fixed aggregate capital, idiosyncratic wedges at the establishment level can depress aggregate TFP by 30-50 percent by pushing resources toward low-productivity establishments. This project builds directly on that insight.

To contextualize this thesis, this review synthesizes four strands of literature: (i) misallocation and aggregate TFP, (ii) heterogeneous-firm models with entry and exit, (iii) empirical and structural work linking wedges to data, and (iv) dynamic policy evaluation and the role of adjustment frictions.

2.1 Misallocation as a Source of TFP Differences

Restuccia and Rogerson (2008) show that when establishments face heterogeneous output distortions, optimal plant size scales with $(1 - \tau)^{1/(1-\alpha-\gamma)} s^{1/(1-\alpha-\gamma)}$, so distortions that increase with productivity induce substantial aggregate losses. Their quantitative experiments demonstrate that even moderate wedges, if correlated with productivity, can reduce aggregate TFP by one-third or more.

A large empirical literature reinforces this mechanism. Using plant-level data for China and India, Hsieh and Klenow (2009) document substantial dispersion in revenue productivity and argue that a large share of cross-country TFP gaps can be explained by misallocation of inputs across establishments. Similarly, Bartelsman et al. (2013) show that differences in allocation and selection across countries are closely linked to institutional and regulatory environments. These findings establish misallocation as an important factor in cross-country productivity variation.

Furthermore, Baqaee and Farhi (2020) develop a general-equilibrium theory of misallocation that shows how micro-level distortions interact with ideas like input-output linkages and nonlinear aggregation to produce potentially much larger aggregate TFP losses than the representative-agent or Cobb-Douglas benchmarks would suggest. Their framework clarifies that misallocation effects depend not only on the dispersion of wedges, but also on the curvature of production functions and the structure of the input-output network. They demonstrate in their environment that the effects of idiosyncratic distortions can be magnified. Although this project does not model this, the existence of such propagation further motivates the study of the transition paths between steady-states.

Further, a set of quantitative studies embed misallocation into environments with ad-

ditional frictions. Buera and Shin (2013) show that financial frictions impede the ability of high-productivity establishments to scale, making misallocation persistent. Buera et al. (2014) study how credit shocks transmit through both capital and labor markets. Midrigan and Xu (2014) provide evidence from plant-level data that financing frictions are a major source of input misallocation in developing economies. All of these papers relate to the central insight of Restuccia and Rogerson (2008): that wedge dispersion at the micro level generates large aggregate consequences.

2.2 Firm Dynamics, Entry/Exit, and Aggregation

The theoretical backdrop of the RR environment came from the heterogeneous-firm model of Hopenhayn (1992). This framework features idiosyncratic productivity, fixed operating costs, endogenous exit, and a free-entry condition that pins down equilibrium profits. Hopenhayn and Rogerson (1993) extend this to analyze policy interventions in a general-equilibrium setting, emphasizing how regulatory wedges affect entry and exit.

These contributions establish the microeconomic structure that is then aggregated: each period, establishments solve static profit maximization problems. Entry is determined by a zero-profit condition, and the invariant distribution of firms characterizes the aggregate allocation.

A separate literature introduces dynamic frictions into heterogeneous-firm production. For example, Khan and Thomas (2008) study capital adjustment costs in an environment with firm-level heterogeneity and show that misallocation interacts with investment frictions to slow convergence. Their work demonstrates how adjustment dynamics matter for welfare and aggregate outcomes.

2.3 Empirical Applications and Policy Interpretation

The empirical literature often interprets cross-sectional wedge dispersion as evidence of some policy issue like regulatory burdens, credit favoritism, size-dependent policies, or restricted entry. The implication is that reforms, whether abrupt or gradual, anticipated or unanticipated, should have dynamic consequences. Removing distortions does not instantaneously deliver the implied TFP gain because capital adjusts slowly and because the composition of entering versus incumbent firms evolves only gradually.

There is less misallocation literature on dynamic questions. This thesis aims to explore this area. Some questions that could be explored in this vein are: (i) the timing of output and consumption gains following reform, (ii) the welfare implications of alternative reform sequences, (iii) how entry and exit respond when distortions change over time, and (iv) how

unanticipated versus foreseen reforms generate different transitional behavior.

The Python notebooks associated with this thesis explore numerical implementations that could explore such questions.

3 Baseline Restuccia-Rogerson Model (With Preferences Specified)

This section rewrites Sections 2-3 of RR in compact notation, keeping all the blocks that matter for defining the steady state, especially: (i) the entry cost c_e , (ii) the operating cost c_f , (iii) the exogenous exit rate λ , (iv) the fact that the household problem pins down the rental rate r , and (v) the fact that the free-entry condition pins down the wage w .

3.1 Household

There is an infinitely lived representative household with preferences

$$\sum_{t=0}^{\infty} \beta^t u(C_t), \quad u(C) = \begin{cases} \frac{C^{1-\sigma}}{1-\sigma}, & \sigma \neq 1, \\ \log C, & \sigma = 1, \end{cases} \quad 0 < \beta < 1,$$

owning the capital stock K_t and supplying one unit of labor inelastically each period. In the steady state, RR impose the usual Euler equation for capital to obtain a fixed user cost of capital,

$$r = \frac{1}{\beta} - (1 - \delta), \tag{1}$$

so the rental rate is not a free variable. This feature of the RR environment is important for equilibrium determination. Because the household Euler equation fixes the rental rate of capital as a function of preference parameters, the interest rate does not adjust to clear markets. It is fixed by (β, δ) . This is critical as when I later solve the firm and entry problems in the replication, r is already known. Instead, equilibrium adjustment occurs through the wage. The wage must therefore move so that the expected value of entry equals the entry cost, which means the free-entry condition pins down the equilibrium wage in the steady state.

3.2 Technology at the establishment level

A producing establishment with idiosyncratic productivity $s > 0$ and inputs (k, n) produces

$$f(s, k, n) = sk^\alpha n^\gamma, \quad \alpha, \gamma \in (0, 1), \quad \alpha + \gamma < 1.$$

There is a per-period fixed operating cost $c_f \geq 0$ (in units of output) that must be paid for the establishment to stay active. There is an exogenous probability of death $\lambda \in (0, 1)$ for each active establishment. There is also an entry cost $c_e > 0$ that must be paid to draw (s, τ) from a policy-induced joint distribution $g(s, \tau)$.

3.3 Policy distortions

RR let each establishment face an idiosyncratic output wedge τ so that revenue is $(1 - \tau)f(s, k, n)$, where τ may be positive (tax) or negative (subsidy), and where τ is drawn jointly with s . Let $g(s, \tau)$ denote the joint pdf over (s, τ) induced by the policy rule $P(\tau|s)$ and the exogenous productivity draw $h(s)$. Changing distortions is changing g .

3.4 Incumbent establishment problem

Given prices (r, w) and a draw (s, τ) , an incumbent establishment solves

$$\pi(s, \tau) = \max_{k, n \geq 0} \{ (1 - \tau)sk^\alpha n^\gamma - rk - wn - c_f \}. \quad (2)$$

The FOCs are

$$r = (1 - \tau)s\alpha k^{\alpha-1}n^\gamma, \quad (3)$$

$$w = (1 - \tau)s\gamma k^\alpha n^{\gamma-1}. \quad (4)$$

Dividing gives

$$\frac{k}{n} = \frac{\alpha}{\gamma} \frac{w}{r},$$

so the plant-level capital to labor ratio depends on prices but not on the wedge. Wedges only scale the establishment. Substituting back yields closed-form demands

$$k(s, \tau; r, w) = \kappa_k(r, w)((1 - \tau)s)^{\frac{1}{1-\alpha-\gamma}}, \quad n(s, \tau; r, w) = \kappa_n(r, w)((1 - \tau)s)^{\frac{1}{1-\alpha-\gamma}}, \quad (5)$$

for explicit $\kappa_k, \kappa_n > 0$.¹

¹These are the same objects the numerical code constructs as `kbar` and `nbar`.

Because the establishment survives with probability $1 - \lambda$ each period and discounts the future at the household rate, its value is

$$W(s, \tau) = \frac{\pi(s, \tau)}{1 - \rho}, \quad \rho \equiv \frac{1 - \lambda}{1 + R}, \quad R \equiv r - \delta = \frac{1}{\beta} - 1. \quad (6)$$

3.5 Entry and the zero-profit condition

A potential entrant pays c_e and then draws (s, τ) from $g(s, \tau)$. After observing (s, τ) the entrant decides whether to operate. Let $x(s, \tau) \in \{0, 1\}$ denote this decision. The expected value of entry is

$$W^e = \sum_{s, \tau} x(s, \tau) W(s, \tau) g(s, \tau) - c_e. \quad (7)$$

With free entry, $W^e = 0$. Since $W(s, \tau)$ is decreasing in w , this zero-profit condition pins down the wage w in equilibrium.

3.6 Invariant distribution of active establishments

Let $\mu(s, \tau)$ denote the steady-state mass of active establishments of type (s, τ) . For a mass E of entrants each period,

$$\mu(s, \tau) = E \cdot \frac{x(s, \tau)}{\lambda} g(s, \tau), \quad (8)$$

because $Eg(s, \tau)$ potential producers arrive, a fraction $x(s, \tau)$ operates, and producers survive $1/\lambda$ periods on average.

3.7 Aggregation and market clearing

Aggregate labor, capital, and output are

$$N(r, w) = \sum_{s, \tau} n(s, \tau; r, w) \mu(s, \tau), \quad (9)$$

$$K(r, w) = \sum_{s, \tau} k(s, \tau; r, w) \mu(s, \tau), \quad (10)$$

$$Y(r, w) = \sum_{s, \tau} [s k(s, \tau; r, w)^\alpha n(s, \tau; r, w)^\gamma - c_f] \mu(s, \tau). \quad (11)$$

Labor supply is 1, so $N(r, w) = 1$ determines E given (r, w) :

$$E = \frac{1}{\sum_{s, \tau} n(s, \tau; r, w) \frac{x(s, \tau)}{\lambda} g(s, \tau)}.$$

3.8 Measured TFP

Following RR, define

$$A(g) \equiv \frac{Y(r, w; g)}{K(r, w; g)^\alpha N(r, w; g)^\gamma}. \quad (12)$$

With $N = 1$, this simplifies to $A(g) = Y/K^\alpha$. Different policies g imply different $A(g)$.

3.9 Equilibrium

A steady-state competitive equilibrium with entry is a tuple

$$(r, w, E, \mu, \{k(s, \tau), n(s, \tau), x(s, \tau)\}_{s, \tau})$$

such that

- (i) r satisfies (1)
- (ii) (k, n) solve (2) for all (s, τ)
- (iii) the free-entry condition (7) holds and determines w
- (iv) μ satisfies (8)
- (v) markets clear: $N = 1$, $K = \sum k\mu$, and the resource constraint (including c_f and $c_e E$) holds
- (vi) The government budget constraint holds (tax revenue redistributed lump sum)

4 Calibration

RR calibrate to the U.S. undistorted economy. I follow their choices in the replication notebook.

- **Time period:** 1 year.
- **Preferences:** $\beta = 0.96$ to match a 4% real return.

- **Technology:** $\alpha + \gamma = 0.85$ from plant-level evidence. I split according to income shares so $\alpha = 0.283$ and $\gamma = 0.567$.
- **Depreciation:** $\delta = 0.08$ so that $I/Y \approx 0.20$ and $K/Y \approx 2.3$.
- **Exit rate:** $\lambda = 0.10$ per year to match U.S. job destruction.
- **Entry cost:** $c_e = 1$ (normalization).
- **Operating cost:** $c_f = 0$ in the baseline.
- **Productivity support:** $s \in [1, 3.98]$ on a log grid with 100 points, matching the U.S. establishment size distribution.

Table 1: Benchmark calibration to U.S. data

Parameter	Value	Target
α	0.283	Capital income share
γ	0.567	Labor income share
β	0.960	Real rate of return
δ	0.080	Investment/output ratio
λ	0.100	Annual exit rate
ce	1.000	Normalization
cf	0.000	Benchmark case
S range	[1.0, 3.98]	Relative establishment sizes

The replication preserves the full structure of the RR model and reproduces all relative statistics and counterfactual TFP losses. Minor differences in aggregate values K , Y , S arise from numerical differences like the density of the grid.

5 Numerical Implementation

This section documents the numerical implementation used to replicate RR. The original work is done in MATLAB, but this replication (and the extension) is in Python.

5.1 Grids and productivity distribution

I construct a log-spaced grid for s with 100 points on $[1, 3.98]$. Given the analytical solution (5), establishment size is proportional to $s^{1/(1-\alpha-\gamma)}$, so I use s to match the U.S. establishment distribution (Table 2 in RR). When the file `establishment_dist.txt` from the original paper is available, the code spreads observed mass over the s grid. Otherwise, it uses the three-bin fallback (share < 5 , 5-49, and ≥ 50 workers).

5.2 Distortions

Distortions are implemented by building (s, τ) matrices. I allow:

- (a) Uncorrelated (i.i.d.) distortions: a fixed fraction is subsidized, a fixed fraction is taxed, the remainder is undistorted
- (b) Correlated distortions: low- s establishments are subsidized and high- s establishments are taxed, or vice versa

5.3 Solving for wages and the invariant distribution

For any given distortion configuration the code:

- (i) takes (β, δ) to compute r
- (ii) guesses a wage w
- (iii) solves plant problems and computes plant values $W(s, \tau)$
- (iv) imposes the free-entry condition to update w via bisection
- (v) once w is found, uses labor-market clearing to find the mass of entrants E
- (vi) constructs the invariant distribution using (8).

5.4 Outputs

For every policy configuration I compute aggregates

$$(Y, K, M, A, E, w)$$

and distributional statistics (shares of establishments, output, labor, capital by employment bin). These are the objects underlying Tables 1-9 in the original paper.

6 Replication of RR Distortion Experiments

What follows are the outputs from the original experiments. They will be useful for the purposes of contextualizing the extensions.

6.1 Benchmark economy

The benchmark with no distortions (all mass on $\tau = 0$) yields:

- Aggregate output $Y = 5.400$
- Aggregate capital $K = 12.575$
- Mass of establishments $M = 5.956$
- Measured TFP $A = 2.636$
- Wage $w = 3.060$

Table 2: Distribution statistics of benchmark economy

	Share of establishments	Output share	Labor share	Capital share	Avg employment
< 5	0.55	0.08	0.08	0.08	0.04
5 to 49	0.39	0.34	0.34	0.34	0.24
≥ 50	0.06	0.58	0.58	0.58	2.82

6.2 Uncorrelated output distortions

RR show that even uncorrelated distortions, where establishments are randomly taxed or subsidized, can lower aggregate TFP substantially once I preserve aggregate capital by choosing a suitable subsidy rate. My implementation reproduces this:

Table 3: Uncorrelated output distortions (50/50 taxed/subsidized): aggregates

	Rel Y	Rel TFP	Rel E	Rel M	Rel w	Y_s/Y	S/Y	τ_{au_s}
τ_{au_t}								
0.100000	0.98	0.98	1.00	1.00	1.00	0.72	0.02	0.06
0.200000	0.96	0.96	1.00	1.00	1.00	0.85	0.05	0.09
0.300000	0.93	0.93	1.00	1.00	1.00	0.93	0.07	0.10
0.400000	0.92	0.92	1.00	1.00	1.00	0.97	0.09	0.11

The key observations are:

- For modest taxes $\tau_t = 0.1$, output and TFP fall by almost 2%.
- For larger taxes $\tau_t = 0.4$, the fall in TFP is about 8-9%, even though capital is preserved.

Table 4: Relative TFP under uncorrelated idiosyncratic output distortions

	0.1	0.2	0.3	0.4
Fraction taxed (percent)				
90	0.92	0.84	0.78	0.74
80	0.95	0.89	0.84	0.81
60	0.98	0.94	0.91	0.89
50	0.98	0.96	0.93	0.92
40	0.99	0.97	0.95	0.94
20	1.00	0.99	0.98	0.97
10	1.00	0.99	0.99	0.99

- A large fraction of output is produced by subsidized plants in these experiments, explaining the positive fiscal cost S/Y .
- Relative TFP also falls as the portion of establishments being taxed grows.

As in the original, I vary the fraction of taxed establishments and show that TFP losses are increasing in the tax rate and in the fraction of establishments exposed to the tax.

6.3 Correlated output distortions

When taxes fall disproportionately on high-productivity or large establishments, the losses are much larger. The notebook’s run shows, for example, that with 50% of low- s firms subsidized and the rest taxed,

- a 10% tax on the taxed group lowers TFP to about 83% of the benchmark
- a 40% tax lowers TFP to about 54% of the benchmark.

This establishes that correlated misallocation is significantly more costly than i.i.d. misallocation.

Table 5: Aggregate variables under correlated idiosyncratic output distortions (subsidize low s)

τ_t	Rel Y	Rel TFP	Rel E	Rel M	Rel w	Y_s/Y	S/Y	τ_s
0.100000	0.83	0.83	1.00	1.00	1.00	0.34	0.21	0.80
0.200000	0.68	0.68	1.00	1.00	1.00	0.59	0.46	0.92
0.300000	0.59	0.59	1.00	1.00	1.00	0.78	0.68	0.96
0.400000	0.54	0.54	1.00	1.00	1.00	0.90	0.84	0.98

Table 6: Relative TFP under correlated idiosyncratic output distortions (subsidize low s)

	0.1	0.2	0.3	0.4
Fraction taxed (percent)				
90	0.81	0.68	0.58	0.52
80	0.84	0.70	0.62	0.57
60	0.88	0.77	0.69	0.65
50	0.90	0.80	0.73	0.69
40	0.92	0.82	0.76	0.72
20	0.95	0.89	0.84	0.81
10	0.97	0.92	0.88	0.86

6.4 Exempt-from-tax experiments

The code also reproduces an experiment in which a fraction of establishments is exempt from a high tax. As the exempt share rises from 10% to 90%, output and TFP rise monotonically toward the benchmark.

Table 7: Taxing all but exempt establishments (output base, $\tau_t=0.4$)

	Rel Y	Rel TFP	Rel E	Rel w	Rel K	Y_s/Y
Exempt (percent)						
10	0.67	0.78	0.58	0.58	0.58	0.67
30	0.78	0.85	0.74	0.74	0.74	0.89
50	0.86	0.90	0.84	0.84	0.84	0.95
70	0.92	0.95	0.91	0.91	0.91	0.98
90	0.98	0.98	0.97	0.97	0.97	0.99

6.5 Capital and labor taxes

RR also consider wedges on capital and labor. My implementation builds tax matrices for these bases and, when desired, chooses subsidies to keep aggregate capital at its benchmark level. The same pattern emerges: correlated wedges on capital services generate larger losses than uncorrelated ones. Wage-based distortions have different interpretations but similar qualitative effects on TFP.

Table 8: Idiosyncratic distortions to capital rental rates

		Rel Y	Rel TFP	Rel E	Y_s/Y	S/Y	τ_s
Case	τ_t						
Uncorrelated	0.500000	0.97	0.97	0.97	0.74	0.01	0.14
	1.000000	0.95	0.95	0.95	0.83	0.02	0.15
Correlated	0.500000	0.89	0.89	0.89	0.33	0.03	0.51
	1.000000	0.82	0.82	0.82	0.46	0.06	0.51

Table 9: Idiosyncratic distortions: output vs. wage base ($\tau_t = -\tau_s = 0.5, 50/50$)

	Rel Y	Rel TFP	Rel K	Rel E	Rel w	Y_s/Y	S/Y
Output base	1.14	0.98	1.70	1.70	1.70	1.00	0.50
Wage base	0.84	0.89	0.84	0.84	1.67	0.98	0.55

This concludes the replication.

7 Dynamic Extensions: Overview

This section provides a high-level conceptual overview of the two dynamic extensions developed in the thesis before turning to the formal theory.

7.1 Motivation for Dynamic Embedding

The static RR environment identifies a map:

$$g \mapsto A(g),$$

where g is a distortion schedule and $A(g)$ is the resulting measured TFP. If distortions change over time, this suggests a sequence $\{A_t\}$ that interacts with the household Euler equation governing capital accumulation. However, RR do not model this transition. Doing so allows us to better understand the impacts of policy reform on welfare as well as misallocation as a whole.

7.2 Reduced-Form Dynamic Embedding

I began by formulating an extension that re-solves the RR block each period using that period's distortions. The resulting measured TFP A_t feeds into a CRRA neoclassical growth

model. This environment was only partially dynamic: firms are static each period, and the only dynamic state is the capital stock. It isolates the interaction between misallocation and capital accumulation without introducing firm investment or equity pricing. This approach was insufficient, but the results can be seen in the replication notebook in the appendix.

7.3 Fully Dynamic Two-Asset Model

The second dynamic extension makes firm accumulation explicit. The firm distribution evolves through entry and exit, and firms become a tradable asset. Households choose next period's capital and equity holdings, and firm mass becomes a dynamic state variable alongside aggregate capital. Policy changes have immediate effects on firm value and therefore on household portfolio choices and entry incentives. This extension is the focus of the rest of this work.

8 Overview and Main Objects

I now develop a dynamic general equilibrium model with:

- (i) a representative household with forward-looking consumption-savings and portfolio choice,
- (ii) forward-looking incumbent firms with dynamic capital adjustment,
- (iii) persistent idiosyncratic wedges as in Restuccia-Rogerson (RR),
- (iv) exogenous firm death at rate $\lambda \in (0, 1)$,
- (v) endogenous entry determined by household optimality (free-entry with complementarity),
- (vi) a time-varying cross-sectional distribution of firms μ_t .

The main difference relative to a single-asset formulation is that the household chooses holdings of two assets:

- (A1) a one-period risk-free asset (or physical-capital claim) with gross payoff $1 + r_{t+1} - \delta$,
- (A2) an equity claim to the firm sector (a claim to the dividend stream of incumbent firms), priced by an endogenous equity price P_t and yielding a gross equity return R_{t+1}^e .

The resulting equilibrium includes two Euler equations (one for each asset) and an explicit definition of dividends and the equity return in terms of firm outcomes and the evolving distribution.

Throughout, I focus on perfect foresight transitions driven by a deterministic policy path for the entrant-type distribution $g_t(s, \tau)$. All expectations are essentially cosmetic under perfect foresight. I retain $\mathbb{E}_t[\cdot]$ notation when it clarifies economic meaning.

9 Environment, Firm Types, and Wedges

Time is discrete, $t = 0, 1, 2, \dots$. There is a single final good used for consumption, investment, and other costs. Labor supply is inelastic and normalized to one.

Firm heterogeneity. A firm is characterized by a type (s, τ) and installed capital $k \geq 0$:

- productivity $s \in S \subset \mathbb{R}_+$,
- output wedge $\tau \in T \subset (-\infty, 1)$,
- capital $k \in \mathbb{R}_+$.

Following RR, (s, τ) is fixed for the life of a firm. New entrants draw $(s, \tau) \sim g_t(s, \tau)$. The distribution g_t may vary deterministically over time, which constitutes a policy reform.

Exogenous firm death. Each period, incumbents die with probability $\lambda \in (0, 1)$, independent of states. Survival probability is $1 - \lambda$.

Technology and wedge interpretation. An incumbent firm with state (s, τ, k) chooses labor $n \geq 0$ to produce

$$y = f(s, k, n) \equiv sk^\alpha n^\gamma, \quad \alpha, \gamma \in (0, 1), \quad \alpha + \gamma < 1. \quad (13)$$

The wedge distorts revenues as in RR:

$$\text{effective revenue} = (1 - \tau) y. \quad (14)$$

Capital depreciates at rate $\delta \in (0, 1)$.

Prices. The competitive wage is w_t and the rental rate of capital is r_t . The equity price is P_t (discussed later).

10 Firm Block

10.1 Static labor demand

Fix t and a firm in state (s, τ, k) . Given prices (w_t, r_t) , the firm chooses labor n to maximize static operating profit:

$$\pi_t(s, \tau, k) = \max_{n \geq 0} \left\{ (1 - \tau)sk^\alpha n^\gamma - w_t n \right\} - r_t k - c_f, \quad (15)$$

where $c_f \geq 0$ is a fixed operating cost.

Assuming an interior solution, the labor FOC is:

$$(1 - \tau)sk^\alpha \gamma n^{\gamma-1} = w_t. \quad (16)$$

Then labor demand is

$$n_t(s, \tau, k) = \left(\frac{\gamma(1 - \tau)s}{w_t} \right)^{\frac{1}{1-\gamma}} k^{\frac{\alpha}{1-\gamma}}. \quad (17)$$

Maximized operating profit. Using the Cobb-Douglas envelope $w_t n_t = \gamma(1 - \tau)sk^\alpha n_t^\gamma$, maximized operating profit is

$$\pi_t(s, \tau, k) = (1 - \gamma)(1 - \tau)sk^\alpha n_t(s, \tau, k)^\gamma - r_t k - c_f. \quad (18)$$

10.2 Dynamic capital choice with adjustment costs

Capital evolves as

$$k' = (1 - \delta)k + i, \quad k' \geq 0, \quad (19)$$

where i is investment in units of the final good. Firms may face convex adjustment costs $\Phi(i, k) \geq 0$, with $\Phi(0, k) = 0$.

10.3 Firm value and Bellman equation

Let $V_t(s, \tau, k)$ denote the value of an incumbent alive at the start of period t with state (s, τ, k) . Because firms are owned by the household, continuation payoffs are discounted by the household's stochastic discount factor (SDF) $m_{t,t+1}$, defined in Section 11.5.

The firm chooses $k' \geq 0$ to solve

$$V_t(s, \tau, k) = \max_{k' \geq 0} \left\{ \pi_t(s, \tau, k) - i - \Phi(i, k) + (1 - \lambda) \mathbb{E}_t [m_{t,t+1} V_{t+1}(s, \tau, k')] \right\}, \quad (20)$$

where $i = k' - (1 - \delta)k$. The policy function is denoted $k'_t(s, \tau, k)$.

Remark (no endogenous exit). Firms do not choose exit; exit occurs only via exogenous death λ .

11 Household Block

11.1 Preferences and timing

The representative household has preferences

$$\sum_{t=0}^{\infty} \beta^t u(C_t), \quad u(C) = \begin{cases} \frac{C^{1-\sigma}}{1-\sigma}, & \sigma \neq 1, \\ \log C, & \sigma = 1, \end{cases} \quad (21)$$

with discount factor $\beta \in (0, 1)$ and risk aversion $\sigma > 0$.

11.2 Assets and returns

The household trades two assets:

(H1) **Risk-free asset** A_{t+1} purchased at t in units of the final good. Its gross payoff is

$$R_{t+1}^f \equiv 1 + r_{t+1} - \delta. \quad (22)$$

(H2) **Equity claim to the firm sector.** Let Q_{t+1} be equity holdings chosen at t . One unit of equity has price P_t at time t and yields (i) a dividend D_{t+1} at time $t + 1$ and (ii) a continuation value P_{t+1} . Thus the gross equity payoff is

$$R_{t+1}^e \equiv \frac{D_{t+1} + P_{t+1}}{P_t}. \quad (23)$$

11.3 Household budget constraint

Let T_t denote lump-sum transfers (government closure defined later), and let wage income be w_t (labor supply is 1). The household's period- t budget constraint is:

$$C_t + A_{t+1} + P_t Q_{t+1} = w_t + R_t^f A_t + (D_t + P_t) Q_t + T_t, \quad (24)$$

given initial holdings (A_0, Q_0) . Equation (24) is written so that asset payoffs at t (from holdings chosen at $t - 1$) finance current consumption and new asset purchases.

Equity supply. The household owns the firm sector, so the supply of equity is normalized to one:

$$Q_t = 1 \quad \text{for all } t. \quad (25)$$

In equilibrium, the household holds the entire equity claim. The Euler equation for equity is meaningful and pins down P_t .

11.4 Household dynamic problem and Bellman equation

Let $\mathcal{F}_t$ denote the information set at t . The household chooses processes $\{C_t, A_{t+1}, Q_{t+1}\}_{t \geq 0}$ to maximize (21) subject to (24) and standard transversality conditions.

A recursive formulation uses wealth $W_t \equiv R_t^f A_t + (D_t + P_t)Q_t$ plus income $w_t + T_t$ as resources. Define the value function

$$\mathcal{V}_t(A_t, Q_t) = \max_{C_t, A_{t+1}, Q_{t+1}} \left\{ u(C_t) + \beta \mathbb{E}_t[\mathcal{V}_{t+1}(A_{t+1}, Q_{t+1})] \right\} \quad (26)$$

subject to (24).

11.5 Euler equations and SDF

The household's first-order conditions imply:

$$u'(C_t) = \beta \mathbb{E}_t[u'(C_{t+1})R_{t+1}^f], \quad (27)$$

$$u'(C_t)P_t = \beta \mathbb{E}_t[u'(C_{t+1})(D_{t+1} + P_{t+1})]. \quad (28)$$

Define the stochastic discount factor

$$m_{t,t+1} \equiv \beta \frac{u'(C_{t+1})}{u'(C_t)}. \quad (29)$$

Then (27)–(28) can be written as asset pricing conditions:

$$1 = \mathbb{E}_t[m_{t,t+1}R_{t+1}^f], \quad (30)$$

$$P_t = \mathbb{E}_t[m_{t,t+1}(D_{t+1} + P_{t+1})]. \quad (31)$$

Euler-implied return and consistency with firm pricing. Along the transition, I impose perfect foresight and compute the intertemporal marginal rate of substitution

$$m_{t,t+1} \equiv \beta \frac{u'(C_{t+1})}{u'(C_t)}.$$

The corresponding one-period gross risk-free return is therefore pinned down by the household Euler equation:

$$1 = \mathbb{E}_t[m_{t,t+1}(1 + r_{t+1})], \quad \text{so under perfect foresight} \quad 1 + r_{t+1} = \frac{1}{m_{t,t+1}}. \quad (32)$$

For CRRA utility $u(C) = \frac{C^{1-\sigma}}{1-\sigma}$, this implies

$$1 + r_{t+1} = \frac{1}{\beta} \left(\frac{C_{t+1}}{C_t} \right)^\sigma.$$

In equilibrium this Euler-implied return must coincide with the marginal product of capital generated by the production side of the economy. Note that using (32) to infer the return sequence makes sense as all privately issued claims are priced using the same stochastic discount factor $m_{t,t+1}$. Because firms are owned by the representative household, their cash flows and continuation values are discounted by $m_{t,t+1}$, and the implied return sequence is the shadow intertemporal price supporting the transition under perfect foresight. Market clearing (in particular, the asset-market condition equating household net saving with the aggregate capital stock and the firm distribution) ensures that this Euler-implied return is compatible with the aggregate allocation. In this environment, the return sequence is not imposed from an aggregate marginal-product identity - instead it is the intertemporal price consistent with the household's Euler equation, while firm values are priced using the same SDF.

12 Entry as a Household Portfolio Decision

Entry creates new firms at cost c_e per entrant. Entrants draw $(s, \tau) \sim g_t(s, \tau)$ and begin at $t + 1$ with initial capital k_0 . Because incumbents die with probability λ , a new entrant survives into $t + 1$ with probability $(1 - \lambda)$.

There are at least two equivalent ways we could think about entry in this two-asset economy:

12.1 Entry as equity issuance at price P_t

We could view new firms as additional equity units issued by a competitive “entry sector,” then the household can buy newly issued equity at price P_t , and proceeds fund the real resource cost c_e . In equilibrium with a representative household and a normalized equity supply, this representation is without loss; the entry condition becomes an asset pricing con-

dition equating marginal cost to marginal value. I choose another representation explained below.

12.2 Free-entry condition in SDF form

Let the *value of an entrant* at t be the SDF-discounted value of an incumbent next period:

$$\text{PV}_t^{\text{entry}} \equiv (1 - \lambda) \mathbb{E}_t \left[m_{t,t+1} \mathbb{E}_{(s,\tau) \sim g_t} [V_{t+1}(s, \tau, k_0)] \right]. \quad (33)$$

Free entry (with complementarity) is:

$$E_t \geq 0, \quad c_e \geq \text{PV}_t^{\text{entry}}, \quad E_t \cdot (c_e - \text{PV}_t^{\text{entry}}) = 0. \quad (34)$$

When $E_t > 0$, entry is interior and satisfies equality:

$$c_e = (1 - \lambda) \mathbb{E}_t \left[m_{t,t+1} \mathbb{E}_{g_t} [V_{t+1}(s, \tau, k_0)] \right]. \quad (35)$$

Interpretation. Entry condition (35) is the analogue of the RR free-entry condition, but adapted to a two-asset asset-pricing environment. It states: marginal resource cost of entry equals the SDF-discounted marginal value of a newborn firm (conditional on survival).

13 Distribution Dynamics

Let μ_t be the measure over states (s, τ, k) at the start of period t . Given the policy $k'_t(s, \tau, k)$ define the deterministic transition map

$$T_t(s, \tau, k) \equiv (s, \tau, k'_t(s, \tau, k)). \quad (36)$$

13.1 Entry measure

A mass E_t of entrants draws $(s, \tau) \sim g_t$ and starts at k_0 : for any measurable set $A \subset S \times T \times \mathbb{R}_+$,

$$\mu_t^{\text{entry}}(A) = E_t \int \mathbb{1}\{(s, \tau, k_0) \in A\} g_t(s, \tau) \, ds \, d\tau. \quad (37)$$

13.2 Law of motion

Survival applies to incumbents with probability $(1 - \lambda)$ and survivors transition via T_t . Thus for any measurable set A ,

$$\mu_{t+1}(A) = (1 - \lambda) \int \mathbf{1}\{T_t(x) \in A\} \mu_t(dx) + (1 - \lambda) E_t \int \mathbf{1}\{x_0(z) \in A\} g_t(dz), \quad (38)$$

for all measurable sets A . Here x denotes an incumbent firm state (a productivity, distortion level, and predetermined capital), $T_t(\cdot)$ is the endogenous transition induced by optimal firm policies and idiosyncratic shocks, and $x_0(z)$ denotes the entrant's initial state upon drawing productivity z (with initial capital pinned to the entrant's initial condition).

Remark (entrant survival). The $(1 - \lambda)$ factor multiplying the entrant mass reflects a timing convention: entrants pay the entry cost at t and become operative at $t + 1$ only if they survive the same exogenous death shock. This makes the transition law and the free-entry valuation consistent.

14 Aggregation and Market Clearing

14.1 Labor market

Aggregate labor demand is

$$N_t = \int n_t(s, \tau, k) \mu_t(ds d\tau dk), \quad (39)$$

and labor supply is fixed at one:

$$N_t = 1. \quad (40)$$

14.2 Capital aggregation and asset market clearing

All physical capital is installed inside firms:

$$K_t = \int k \mu_t(ds d\tau dk). \quad (41)$$

Risk-free asset market clearing identifies the household's risk-free holdings with aggregate capital claims. In a representative-agent equilibrium it is standard to set:

$$A_t = K_t. \quad (42)$$

(Equivalently, the risk-free asset is a claim to the net of depreciation with rental rate r_t of installed capital.)

14.3 Output and feasibility

Physical output net of fixed operating costs is

$$Y_t = \int \left(s k^\alpha n_t(s, \tau, k)^\gamma - c_f \right) \mu_t(ds \, d\tau \, dk). \quad (43)$$

Incumbent investment is $i_t(s, \tau, k) \equiv k'_t(s, \tau, k) - (1 - \delta)k$. Aggregate incumbent investment and adjustment cost:

$$I_t^{\text{inc}} = \int i_t(s, \tau, k) \mu_t(ds \, d\tau \, dk), \quad (44)$$

$$\Phi_t^{\text{agg}} = \int \Phi(i_t(s, \tau, k), k) \mu_t(ds \, d\tau \, dk). \quad (45)$$

Goods feasibility is

$$C_t + I_t^{\text{inc}} + \Phi_t^{\text{agg}} + c_e E_t = Y_t. \quad (46)$$

15 Government and Wedge Revenue

Wedges generate tax/subsidy revenue:

$$R_t = \int \tau s k^\alpha n_t(s, \tau, k)^\gamma \mu_t(ds \, d\tau \, dk). \quad (47)$$

Following RR, the government balances its budget each period via a lump-sum transfer T_t :

$$T_t = R_t. \quad (48)$$

Because labor is inelastic, this transfer is not distortionary.

16 Dividends, Equity Price, and Equity Return

We must map firm outcomes to equity cashflows.

16.1 Dividends

Define aggregate *dividends* as the payout from incumbents after financing investment and adjustment costs.

$$D_t \equiv \int \left(\pi_t(s, \tau, k) - i_t(s, \tau, k) - \Phi(i_t(s, \tau, k), k) \right) \mu_t(ds \, d\tau \, dk) - c_e E_t, \quad (49)$$

where π_t already subtracts the rental cost $r_t k$ and fixed cost c_f . The subtraction of $c_e E_t$ reflects that entry costs are paid in goods and reduce distributable resources.

Consistency with feasibility. Combining the definition of D_t with feasibility (46) and household income flows, one can verify that dividends are the residual cashflow from the firm sector that can be distributed to the household without violating goods feasibility.

16.2 Equity pricing

By the household equity Euler equation (31), equilibrium equity prices satisfy

$$P_t = \mathbb{E}_t [m_{t,t+1}(D_{t+1} + P_{t+1})]. \quad (50)$$

Under perfect foresight, this collapses to a deterministic recursion:

$$P_t = m_{t,t+1}(D_{t+1} + P_{t+1}). \quad (51)$$

16.3 Equity return

Given P_t , the gross equity return is

$$R_{t+1}^e = \frac{D_{t+1} + P_{t+1}}{P_t}. \quad (52)$$

The equity Euler condition is equivalent to

$$1 = \mathbb{E}_t [m_{t,t+1} R_{t+1}^e]. \quad (53)$$

Role of exogenous death. Firm death affects dividends and prices through the distribution law of motion (38) and through the firm Bellman equation (20) via the survival factor $(1 - \lambda)$. No endogenous exit is required for equity pricing to be well-defined.

17 Competitive Equilibrium Definition

Fix primitives

$$(\beta, \sigma, \alpha, \gamma, \delta, \lambda, c_e, c_f, k_0),$$

initial conditions (A_0, Q_0, μ_0) and a deterministic policy path $\{g_t\}_{t \geq 0}$.

A perfect-foresight competitive equilibrium is a collection of sequences

$$\{C_t, A_{t+1}, Q_{t+1}; w_t, r_t, P_t, D_t; V_t(\cdot), n_t(\cdot), k'_t(\cdot); \mu_t, E_t, T_t\}_{t \geq 0}$$

such that for all $t \geq 0$:

- E1. Household optimality.** Given prices and transfers, $\{C_t, A_{t+1}, Q_{t+1}\}$ maximizes (21) subject to (24), implying the Euler equations (27) and (28) (equivalently (30) and (31)) and transversality.
- E2. Firm static labor choice.** For each (s, τ, k) , labor $n_t(s, \tau, k)$ solves (15) (equivalently (17)).
- E3. Firm dynamic capital choice.** For each (s, τ, k) , V_t and k'_t solve (20), with discounting via the SDF (29).
- E4. Entry optimality.** Entry E_t satisfies the free-entry complementarity condition (34) (equivalently equality (35) when $E_t > 0$).
- E5. Distribution consistency.** μ_{t+1} evolves according to (??) given μ_t , k'_t , and E_t .
- E6. Market clearing.** Labor clears (40); equity market clears (25); capital aggregates (41) and clears (42); goods feasibility holds (46).
- E7. Dividends and equity pricing.** Dividends satisfy (49) and equity prices satisfy (50). Equivalently, equity returns satisfy (53).
- E8. Government budget.** Transfers satisfy (48).

18 Numerical Implementation

This model requires solving a forward–backward system under perfect foresight. The backward component addresses equity pricing, which depends on the discounted value of future dividends and firm survival. The forward component arises from the household Euler equations determining consumption and savings (in either asset), as well as from the law of motion for firm mass. This section summarizes the structure and logic of this computational approach.

18.1 Backward Recursion: Equity Valuation

Given a terminal date T large enough for the economy to be arbitrarily close to the post-reform steady state, I impose the steady-state equity value P_T consistent with that terminal environment. The no-arbitrage condition

$$P_t = \mathbb{E}_t [m_{t,t+1}(D_{t+1} + P_{t+1})]$$

is then solved backward for $t = T - 1, \dots, 0$.

We can compute the equity price path $\{P_t\}_{t=0}^T$ by backward recursion from the perfect-foresight pricing equation

$$P_t = m_{t,t+1}(D_{t+1} + P_{t+1}),$$

where $m_{t,t+1} = \beta u'(C_{t+1})/u'(C_t)$ and D_{t+1} denotes aggregate dividends. To close the recursion, I impose a terminal condition that approximates the continuation value beyond horizon T by the post-reform steady state. Concretely, I pin a terminal dividend level D^* associated with the post-reform steady state (or the terminal tail of the computed transition), and set P_T equal to the steady-state value that solves

$$P = m^*(D^* + P),$$

where m^* is the steady-state discount factor. Under the standard steady-state normalization with constant consumption (so that $m^* = \beta$), this yields

$$P_T = \frac{\beta}{1 - \beta} D^*. \tag{54}$$

This terminal approximation has negligible influence on the economically relevant portion of the transition provided the horizon T is sufficiently long and the computed allocation has largely converged to the post-reform steady state by the terminal tail.

18.2 Forward Recursion: Capital, Consumption, and Firm Mass

Given prices, the household Euler equations determine consumption and saving decisions such that $Q_t = 1$ is fixed. Given prices, the household Euler equations determine C_t and K_{t+1} , with $Q_{t+1} = 1$ along the transition. Physical capital evolves as we defined before:

$$K_{t+1} = Y_t - C_t - (\text{operating costs}) - c_e E_t,$$

subject to depreciation and any adjustment costs.

Firm mass evolves through:

$$M_{t+1} = (1 - \lambda)M_t + E_t,$$

where entry E_t is determined by the free-entry condition

$$c_e = \mathbb{E}_t [m_{t,t+1} W_{t+1}(s, \tau)],$$

with $W_{t+1}(s, \tau)$ denoting the continuation value of a new firm. Because both W_{t+1} and $m_{t,t+1}$ reflect expectations about future prices, entry responds instantaneously to anticipated changes in any prices.

18.3 Within-Period Plant Allocation (Static RR Block)

For given (K_t, M_t) and distortions at time t , the static RR block determines:

- labor for each (s, τ) ,
- operating profits π_t ,
- dividends D_t ,
- the equilibrium wage w_t satisfying the free-entry condition,
- aggregate output Y_t and TFP.

These objects feed directly into both the household Euler equations and the equity recursion.

18.4 Solving the Full Transition Path

The transition is solved via a damped time-path iteration (TPI) algorithm:

1. Guess initial sequences $\{K_t, M_t\}_{t=0}^T$ consistent with initial conditions.
2. Solve the plant block each period to obtain wages and dividends.
3. Solve the equity pricing recursion backward from T to 0.
4. Solve the household Euler equations forward to update $\{C_t, K_{t+1}, Q_{t+1}\}$.
5. Update the law of motion for M_{t+1} using the entry condition.
6. Form residuals measuring violations of goods market clearing and Euler equations.

7. Update the paths using damped adjustments.

8. Iterate until convergence.

The resulting equilibrium transition includes full time paths for $C_t, K_t, M_t, E_t, P_t, D_t, w_t$, and welfare.

18.5 Details: Transition Algorithm as a Function of Capital and Firm Mass Paths

The key simplifying observation is that a perfect-foresight transition can be computed by guessing only the paths of aggregate capital and the mass of firms,

$$\{K_t, M_t\}_{t=0}^T,$$

and then expressing all other aggregate objects as functions of these two paths and the initial firm distribution μ_0 . This subsection writes out this process and shows the equations that link each object to $\{K_t, M_t\}$.

Step 1: Laws of motion for K_t and M_t

Given a guess $\{K_t, M_t\}_{t=0}^T$ and an exogenous depreciation rate δ and death rate λ , the laws of motion imply unique sequences of aggregate investment and net entry flows. In particular, the capital accumulation equation

$$K_{t+1} = (1 - \delta)K_t + I_t \tag{55}$$

defines aggregate investment as

$$I_t \equiv K_{t+1} - (1 - \delta)K_t, \tag{56}$$

which is a known function of the guessed path $\{K_t\}$.

Similarly, the law of motion for the mass of firms is

$$M_{t+1} = (1 - \lambda)(M_t + E_t), \tag{57}$$

which determines the required mass of entrants as

$$E_t \equiv \frac{M_{t+1}}{1 - \lambda} - M_t. \tag{58}$$

Thus, given $\{K_t, M_t\}$ we also know $\{I_t, E_t\}$.

Step 2: Static RR block given K_t and μ_t

Given the current firm distribution μ_t and aggregate capital K_t , the static Restuccia-Rogerson block determines wages, factor demands, and aggregate output in period t .

Each firm of type (s, τ, k) solves the static labor-demand problem

$$\max_{n_t \geq 0} \pi_t(s, \tau, k) = (1 - \tau) f(s, k, n_t) - w_t n_t - r_t k - c_f, \quad (59)$$

which yields a labor demand function

$$n_t(s, \tau, k) = n(s, \tau, k; w_t, r_t). \quad (60)$$

Aggregating over the distribution μ_t ,

$$N_t(w_t, r_t; \mu_t) \equiv \int n(s, \tau, k; w_t, r_t) \mu_t(ds d\tau dk), \quad (61)$$

and similarly

$$K_t^{\text{inc}} \equiv \int k \mu_t(ds d\tau dk), \quad Y_t(w_t, r_t; \mu_t) \equiv \int y(s, k, n(s, \tau, k; w_t, r_t)) \mu_t(ds d\tau dk). \quad (62)$$

Given the guessed K_t and μ_t , the wage and rental rate are the unique pair (w_t, r_t) that satisfy

$$N_t(w_t, r_t; \mu_t) = 1, \quad (63)$$

$$K_t^{\text{inc}} = K_t, \quad (64)$$

which means that the labor market clears and there is consistency between the aggregate capital carried by incumbents and the guessed path K_t .

Step 3: Dynamic firm problem and implied investment

Given (w_t, r_t) from Step 2 and the household stochastic discount factor $m_{t,t+1}$, each incumbent firm solves the dynamic programming problem

$$V_t(s, \tau, k) = \max_{k' \geq 0} \{ \pi_t(s, \tau, k) - \mathcal{I}(k', k) + (1 - \lambda) m_{t,t+1} V_{t+1}(s, \tau, k') \}, \quad (65)$$

which yields an optimal capital policy

$$k'_t(s, \tau, k) = k'(s, \tau, k; w_t, r_t, m_{t,t+1}). \quad (66)$$

Aggregating across firms, the implied investment of incumbents is

$$I_t^{\text{inc}}(w_t, r_t, m_{t,t+1}; \mu_t) \equiv \int [k'(s, \tau, k; w_t, r_t, m_{t,t+1}) - (1 - \delta)k] \mu_t(ds d\tau dk). \quad (67)$$

Given the implied aggregate investment I_t from the guessed capital path in (56), we require

$$I_t^{\text{inc}}(w_t, r_t, m_{t,t+1}; \mu_t) = I_t, \quad (68)$$

which links the dynamic firm policies to the guessed K_t sequence.

Step 4: Entry and the firm-mass law of motion

The free-entry condition relates the entry cost c_e and the expected value of a new firm. Using the continuation values from (65), the value of entry is

$$\text{EV}_t \equiv (1 - \lambda) \mathbb{E}_{g_t} [m_{t,t+1} V_{t+1}(s, \tau, k_0)], \quad (69)$$

and the complementarity condition is

$$c_e \geq \text{EV}_t, \quad E_t \geq 0, \quad E_t \cdot (c_e - \text{EV}_t) = 0. \quad (70)$$

Given E_t from the guessed mass path via (58), the pair (EV_t, E_t) must satisfy (69)–(70). This pins down the value of entry in a way that is consistent with the guessed mass path $\{M_t\}$.

The full law of motion for the firm distribution can then be written as

$$\mu_{t+1} = \mathcal{T}(\mu_t, k'_t, E_t, g_t, \lambda), \quad (71)$$

and the implied mass of firms is

$$M_{t+1}^{\text{imp}} \equiv \int \mu_{t+1}(ds d\tau dk). \quad (72)$$

The consistency condition for the mass path is

$$M_{t+1}^{\text{imp}} = M_{t+1}, \quad (73)$$

which will be enforced at the TPI fixed point.

Step 5: Aggregate resources and consumption

Given Y_t , I_t^{inc} and E_t , and the aggregate adjustment costs Φ_t^{agg} implied by policies $k'_t(s, \tau, k)$, aggregate dividends are defined as leftover firm cash flow:

$$D_t = Y_t - w_t N_t - r_t K_t^{\text{inc}} - I_t^{\text{inc}} - \Phi_t^{\text{agg}} - c_e E_t. \quad (74)$$

Goods-market feasibility then determines consumption as

$$C_t = Y_t - I_t^{\text{inc}} - \Phi_t^{\text{agg}} - c_e E_t, \quad (75)$$

which is a function of $\{K_t, M_t\}$ through the previously determined objects. The stochastic discount factor is

$$m_{t,t+1} \equiv \beta \frac{u'(C_{t+1})}{u'(C_t)}. \quad (76)$$

Step 6: Equity pricing

With dividends $\{D_t\}$ and discount factors $\{m_{t,t+1}\}$, the aggregate equity price satisfies the backward recursion

$$P_t = m_{t,t+1}(D_{t+1} + P_{t+1}), \quad (77)$$

with terminal condition P_T given by the post-reform steady-state relationship between P , D , and m (see Appendix A.8). Solving (77) yields an equity price path $\{P_t\}$ as a function of the guessed $\{K_t, M_t\}$.

Step 7: Household Euler equations and residuals

Finally, the household Euler equations must hold along the transition. The risk-free Euler equation is

$$u'(C_t) = \beta u'(C_{t+1}) R_{t+1}^f, \quad (78)$$

where R_{t+1}^f is the gross return on the riskless capital asset implied by r_{t+1} , and the equity Euler equation is

$$u'(C_t) P_t = \beta u'(C_{t+1}) (D_{t+1} + P_{t+1}). \quad (79)$$

Given the consumption and price sequences implied by $\{K_t, M_t\}$ through the previous steps, (78)–(79) define Euler residuals

$$R_t^{\text{Euler},f} \equiv u'(C_t) - \beta u'(C_{t+1})R_{t+1}^f, \quad (80)$$

$$R_t^{\text{Euler},e} \equiv u'(C_t)P_t - \beta u'(C_{t+1})(D_{t+1} + P_{t+1}), \quad (81)$$

which must be zero in equilibrium.

Step 8: Fixed point in $\{K_t, M_t\}$

The time–path iteration algorithm uses the mapping described above to construct implied sequences of investment, entry, consumption, prices, and distributions from an initial guess for $\{K_t, M_t\}$. The implied paths generate residuals for

- capital accumulation, via (68),
- the firm mass law of motion, via (73),
- the Euler equations, via $R_t^{\text{Euler},f}$ and $R_t^{\text{Euler},e}$,
- goods feasibility and labor market clearing.

Collecting these residuals into a vector-valued function

$$\mathcal{R}(\{K_t, M_t\}) = 0,$$

TPI updates the guess for $\{K_t, M_t\}$ by a damped fixed-point step until $\mathcal{R}(\{K_t, M_t\})$ is arbitrarily close to zero. At the fixed point, all objects $\{C_t, K_t, M_t, E_t, P_t, D_t, w_t, r_t, \mu_t\}_{t=0}^T$ are jointly determined as functions of the capital and mass paths, and the resulting transition path satisfies all equilibrium conditions.

18.6 Diagnostics

To assess the numerical accuracy of the computed transition, I use time-series residuals for the key equilibrium restrictions that are not imposed mechanically by construction:

- the labor-market clearing residual,

$$\text{Res}_t^L \equiv L_t^{\text{inc}} + L_t^{\text{ent}} - \bar{L},$$

where incumbent and entrant labor demands are aggregated using the current distribution and entrant mass;

- the goods-resource residual,

$$\text{Res}_t^Y \equiv (C_t + I_t + \Phi_t + c_e E_t) - Y_t,$$

which measures violations of the aggregate resource constraint;

- the free-entry (or entry-valuation) residual,

$$\text{Res}_t^E \equiv \frac{\text{PV}_t}{c_e} - 1,$$

where PV_t is the discounted value of a marginal entrant computed using the same pricing kernel as equity.

Across the reported transition, these residuals are small and decay rapidly. In particular, while the maximal goods-resource residual can be modestly larger early in the transition (which is not unexpected for a shooting algorithm) it falls to negligible levels after a short burn-in and remains near zero as the path converges. I also verified that extending the horizon T and tightening tolerances does not materially alter the main economic objects (output, consumption, capital, wages, and welfare changes). In the baseline calibration, the peak goods residual is on the order of $\mathcal{O}(10^{-2})$ (as a share of output) early in the transition and quickly declines; after period $t \geq t_0$ it remains below ε .

19 A Worked Transition: Eliminating Correlated 0.2 Wedges

This section presents the transition dynamics implied by the model when the economy begins in a distorted correlated-wedge steady state. The setting is such that high-productivity entrants disproportionately face a revenue wedge of 0.2 while low-productivity firms receive a corresponding subsidy. At time $t = 0$, there is a permanent reform so that all new entrants face $\tau = 0$. This experiment mirrors the correlated-wedge case studied in *Restuccia and Rogerson (2008)*, but here I characterize the entire transition path under perfect foresight, rather than comparing only steady states. Plots of the transition dynamics appear in Appendix A.

The economy is initialized at the invariant distribution μ_{pre} associated with the distorted correlated-wedge steady state. At $t = 0$ the wedge is permanently removed for all new firms, causing the entrant distribution to jump from g_{pre} to g_{post} , where the latter assigns all mass to the undistorted wedge. The economy then converges gradually to the undistorted steady

state μ_{post} . To highlight the role of transitional dynamics, I compare (i) the full consumption-equivalent welfare change under the transition path and (ii) the welfare difference coming only from the steady-state comparison, as in the original paper.

19.1 Output, TFP, and Production Structure

Aggregate output and productivity along the transition are given by

$$Y_t = \int (sk^\alpha n^\gamma - c_f) \mu_t(ds d\tau dk), \quad \text{TFP}_t = \frac{Y_t}{K_t^\alpha N_t^\gamma},$$

where K_t and N_t are aggregates constructed from μ_t .

Immediately after reform, both Y_t and TFP_t exhibit a small dip. This reflects a short-run disruption as a surge of low-capital entrants arrives and existing firms partially unwind their distorted pre-reform policies. Shortly after, both variables rise sharply. The elimination of correlated distortions allows high-productivity firms to expand and reduces the mass of high-wedge incumbents. Reallocation is rapid, and the economy moves toward a substantially more efficient production structure. Over time, capital deepens and both output and TFP converge smoothly to their higher frictionless steady-state values.

This long-run TFP gain mirrors the magnitude of improvements emphasized in RR. The transition adds some information about what may happen in between.

19.2 Wages, Rental Rates, and Factor Price Adjustment

Factor prices adjust in line with the changing marginal products of inputs.

Wages w_t . Wages exhibit a minor dip at $t = 0$ and then rise steadily to their undistorted steady-state level. The immediate dip occurs because the entry surge depresses the aggregate capital-labor ratio. Then, as reallocation progresses and the share of high-productivity undistorted firms increases, the marginal product of labor increases and wages respond accordingly.

Rental rate r_t . The rental rate displays the characteristic non-monotonic shape. It initially drops sharply on impact because capital is predetermined in the short run and the sudden inflow of entrants increases the demand for labor relative to the existing capital stock, reducing the marginal return on capital momentarily. As TFP rises and capital is reallocated toward productive firms, the marginal product of capital increases, pushing r_t

upward. As capital accumulates, the rental rate declines gradually toward its frictionless user-cost steady state.

These movements show the interaction of capital adjustment frictions, changing firm composition, and the endogenous response of factor demands following distortion removal.

19.3 Entry and the Reallocation Margin

Entry E_t reacts very strongly to the reform. The free-entry condition,

$$(1 - \lambda) \mathbb{E}_t[m_{t,t+1} V_{t+1}(s, \tau, k_0)] = c_e,$$

is immediately violated when all entrants become undistorted and the continuation value of a new plant rises. As a result, the economy experiences a sharp spike in entry at $t = 0$.

This spike plays a central role in shaping the transition. The sudden arrival of many low-capital entrants temporarily depresses factor prices and dividends, and it also raises investment demand. As μ_t gradually realigns to the frictionless mixture of plant types, entry declines to the undistorted steady-state rate. This entry dynamic is the main source of the non-monotonic behavior seen in output and factor prices.

19.4 Equity Prices and Dividends

Firm value satisfies the present-value relation

$$P_t = m_{t,t+1} (D_{t+1} + P_{t+1}),$$

so equity prices respond to both changes in expected profitability and variation in the stochastic discount factor.

On impact, the correlated-wedge reform generates a drop in equity prices. Although the long-run profitability of firms rises, incumbent firms face an immediate erosion of rents due to intense competition from undistorted entrants and increased investment needs. Dividends D_t also fall sharply on impact due to the combination of capital adjustment costs and the depressed marginal product of capital in the early transition.

After this initial disruption, firm values rise rapidly, reflecting higher long-run profitability and the gradual stabilization of discounting. Both P_t and D_t converge from below to their higher steady-state levels.

19.5 Consumption Dynamics and Welfare

Consumption evolves according to the aggregate resource constraint,

$$C_t = Y_t - I_t - \Phi_t - c_e E_t,$$

and thus reflects the path of reallocation costs as well as long-run improvements in productivity. The simulated consumption path exhibits a modest dip at $t = 0$, when entry and investment surge, followed by a steady rise as capital deepens and distortions vanish. Consumption ultimately settles at a level substantially higher than in the distorted steady state.

Welfare differences are computed using a consumption-equivalent measure under CRRA preferences. Let C_{pre} denote distorted steady-state consumption, c_{eq}^{SS} the consumption-equivalent implied by comparing steady states only, and c_{eq}^{trans} the consumption-equivalent along the full transition. As in RR, the steady-state comparison implies large welfare gains from removing correlated wedges. However, once the transition is taken into account, the gains shrink significantly. The initial consumption dip, slow capital adjustment, and discounting, among other things, significantly diminish the measured improvement.

19.6 Summary

The correlated-wedge reform generates a transition characterized by an immediate but short-lived efficiency loss, followed by a strong, sustained rise in production and factor prices. Output, TFP, wages, rental rates, dividends, equity values, and entry all display non-monotonic dynamics arising from capital adjustment frictions and endogenous firm selection. Although the long-run gains from distortion removal are substantial, the transition path reveals that steady-state comparisons overestimate the welfare benefits of such reforms. This suggests incorporating transitional dynamics into policy analysis in heterogeneous-firm environments is important.

20 Discussion and Interpretation

20.1 Immediate Effects Through Equity Valuation and Entry

Because equity value is the discounted sum of future dividends and survival, any policy reform that affects wedges or future profitability produces immediate jumps in P_t . These valuation effects alter household wealth and change the composition of their savings portfolio. Even if physical capital adjusts slowly, firm value adjusts right away in this model.

Entry responds not to current profitability alone but to expected future profitability captured in W_{t+1} . This means that reforms influence the productive sector even before capital responds. A reduction in distortions raises future operating profits and hence P_t , inducing additional entry that raises M_t over time. In this way, entry becomes an amplification channel for reform.

20.2 Dynamic Interaction Between Capital and Firm Mass

Physical capital and firm mass evolve according to distinct adjustment processes:

- K_t evolves gradually through the Euler equation,
- M_t evolves through survival and entry,
- both interact through the plant block, which determines wages and profits.

This interaction can produce transition dynamics in which M_t rises before K_t , creating periods where aggregate labor demand increases but capital remains scarce, which then affects the prices and consumption.

20.3 Potential Role of Expectations and Reform Timing

While not contrasted here, we can hypothesize the effects of expectations and timing based on these findings. Anticipated reforms allow households to shift their portfolios toward equity ahead of time, smoothing the consumption path. Conversely, unanticipated reforms force households to adjust immediately, often inducing sharp increases in investment or equity demand. These differences can be quantitatively significant in welfare terms, and accounting for this is another potential use of studying transition dynamics.

20.4 Welfare Implications

Welfare depends on the entire consumption path. Since consumption is affected by both capital accumulation and firm value, reforms that improve long-run TFP may still involve substantial short-run costs if they induce large investment surges or require temporarily lower consumption. A dynamic framework serves to illustrate these trade-offs across alternative reform sequences.

21 Conclusion

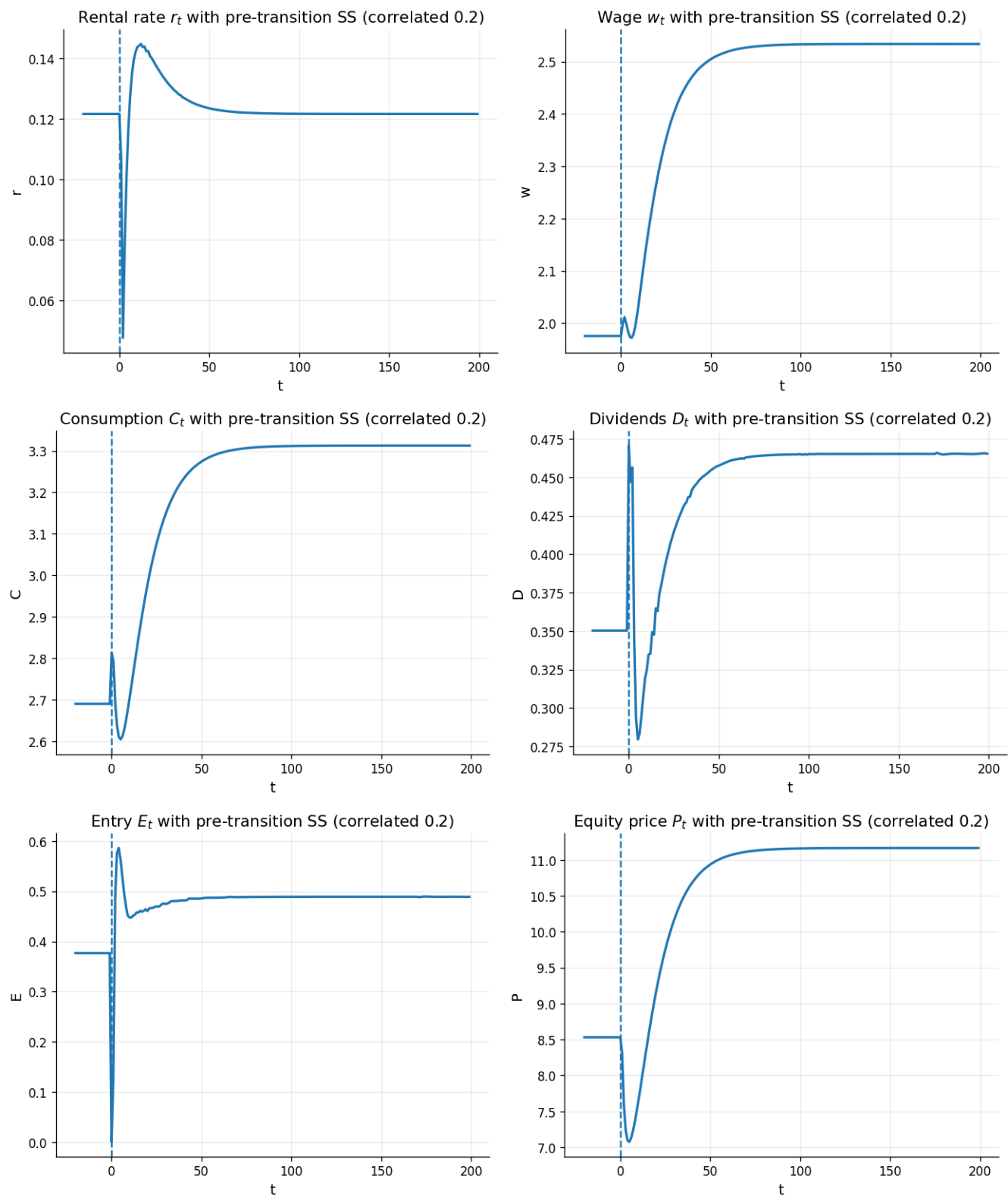
This thesis develops a dynamic, two-asset general equilibrium extension of the Restuccia–Rogerson (2008) misallocation model. By integrating the static plant-level structure of RR into an intertemporal framework with household portfolio choice, firm survival, equity pricing, and endogenous entry, the model illustrates a platform for studying transition dynamics under establishment-level distortions.

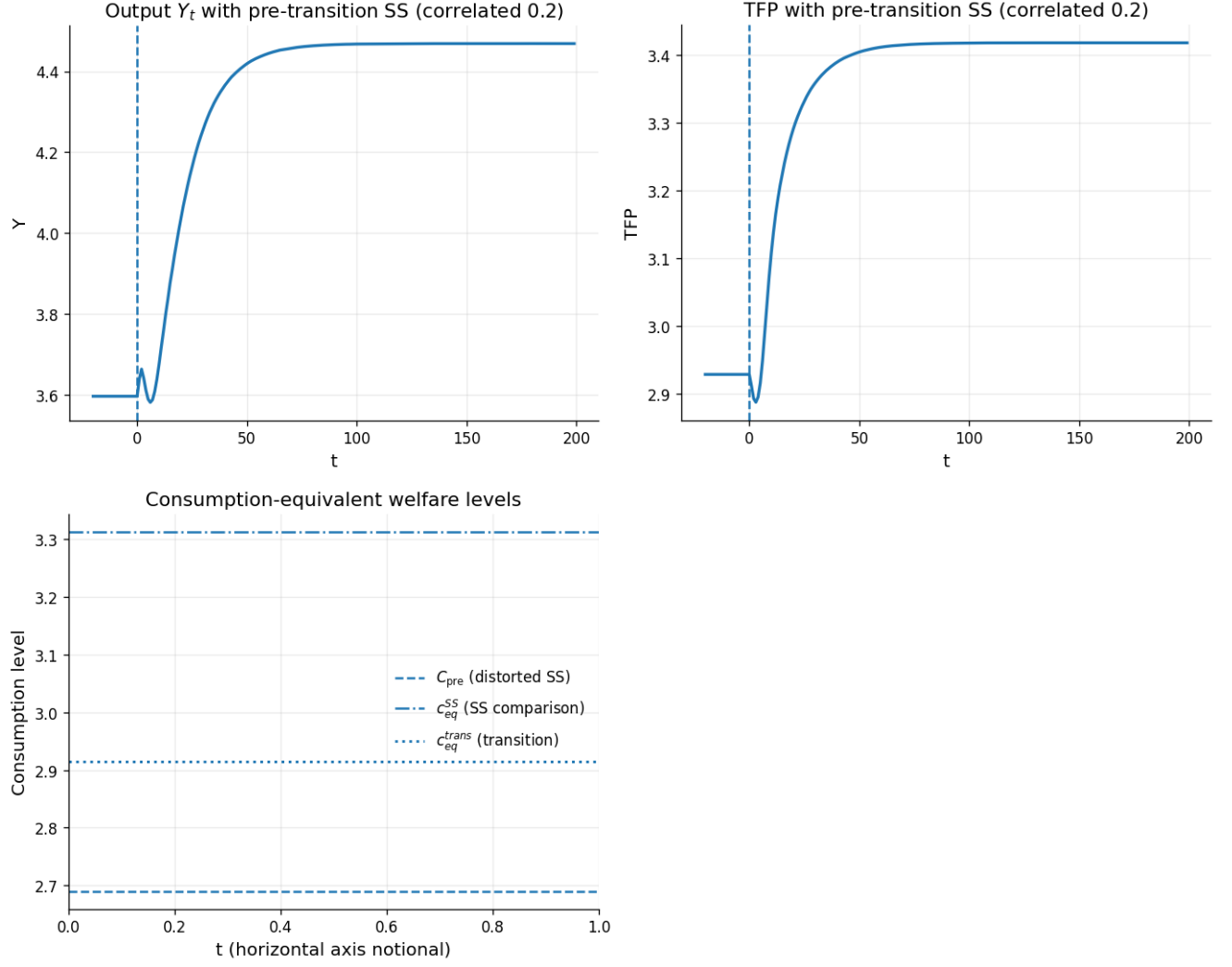
The dynamic extension highlights several channels that may be relevant to understanding misallocation: equity values respond instantaneously to reforms; entry adjusts based on forward-looking profitability; capital and firm mass evolve jointly; and welfare depends on the complex interplay of these adjustments. These features highlight that dynamics are useful for evaluating the design and timing of reforms, and for understanding the transitional costs and benefits that shape welfare.

Future research may extend the model by incorporating heterogeneous-firm investment, transitional selection among incumbents, stochastic policy uncertainty, and richer production linkages. It may also be interesting to examine more realistic and complex policies and their implied transition paths between similar distorted and undistorted steady-states. Further, building a richer model that aims to reflect real data could provide more insight into how misallocation arises and reacts to its environment.

In summary, the benefits of reducing misallocation are not best measured through steady-state comparisons. In the static Restuccia–Rogerson environment, removing distortions generates an immediate improvement in aggregate productivity because resources are instantaneously reallocated toward more productive establishments. In a dynamic setting, these gains appear gradually. Each component of the economy experiences a transition, which is not always monotonic. Because welfare depends on the entire consumption path, these transitional dynamics are important to evaluating a policy. The model developed here illustrates that policies that improve resource allocation can generate long-run welfare gains, but the process of reallocation takes time that should be considered.

A Extension Numerical Results





B Timing Convention

Within each period t , the sequence of events is as follows:

1. The aggregate state (τ_t, μ_t) is given, where τ_t collects policy wedges and μ_t is the distribution of incumbent firms over idiosyncratic productivity and predetermined capital.
2. Incumbent firms produce and pay factor costs, choose investment and next-period capital (subject to adjustment costs), and distribute period- t cash flows.
3. An exogenous survival shock is realized: each incumbent survives into $t + 1$ with probability $(1 - \lambda)$.
4. Entry is chosen at t (the mass E_t) and entrants draw idiosyncratic productivity according to g_t . Entrants start operating at the start of period $t + 1$ if they survive the

same exogenous survival shock, so that only a fraction $(1 - \lambda)$ of the mass E_t enters the operative distribution in period $t + 1$.

This timing is the one implemented in the numerical transition solver; it implies that the entrant mass contributing to μ_{t+1} is $(1 - \lambda)E_t g_t$.

A Mathematical Appendix

This appendix provides derivations and identities underlying the theory in the main text. All objects and timing follow the conventions stated in the main text. Expectations $\mathbb{E}_t[\cdot]$ may be interpreted as degenerate under perfect foresight; the derivations are presented in stochastic form to emphasize general validity.

A.1 Household Problem: Lagrangian, FOCs, SDF, and TVCs

A.1.1 Sequential formulation

The representative household chooses sequences $\{C_t, A_{t+1}, Q_{t+1}\}_{t \geq 0}$ to maximize

$$\max_{\{C_t, A_{t+1}, Q_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(C_t), \quad (82)$$

subject to the per-period budget constraint

$$C_t + A_{t+1} + P_t Q_{t+1} = w_t + R_t^f A_t + (D_t + P_t) Q_t + T_t, \quad t \geq 0, \quad (83)$$

and nonnegativity constraints $C_t \geq 0$ (and potentially borrowing constraints on A_{t+1}).

Here $R_{t+1}^f \equiv 1 + r_{t+1} - \delta$ is the gross risk-free payoff.

A.1.2 Lagrangian and first-order conditions

Let $\{\Lambda_t\}_{t \geq 0}$ be multipliers on (83). The Lagrangian is:

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[u(C_t) + \Lambda_t \left(w_t + R_t^f A_t + (D_t + P_t) Q_t + T_t - C_t - A_{t+1} - P_t Q_{t+1} \right) \right]. \quad (84)$$

FOC for consumption:

$$\frac{\partial \mathcal{L}}{\partial C_t} : \quad \beta^t \left(u'(C_t) - \Lambda_t \right) = 0 \Rightarrow \Lambda_t = u'(C_t). \quad (85)$$

FOC for risk-free asset A_{t+1} :

$$\frac{\partial \mathcal{L}}{\partial A_{t+1}} : \quad -\beta^t \Lambda_t + \beta^{t+1} \mathbb{E}_t [\Lambda_{t+1} R_{t+1}^f] = 0. \quad (86)$$

Using (85),

$$u'(C_t) = \beta \mathbb{E}_t [u'(C_{t+1}) R_{t+1}^f]. \quad (87)$$

FOC for equity Q_{t+1} :

$$\frac{\partial \mathcal{L}}{\partial Q_{t+1}} : \quad -\beta^t \Lambda_t P_t + \beta^{t+1} \mathbb{E}_t [\Lambda_{t+1} (D_{t+1} + P_{t+1})] = 0, \quad (88)$$

which implies

$$u'(C_t) P_t = \beta \mathbb{E}_t [u'(C_{t+1}) (D_{t+1} + P_{t+1})]. \quad (89)$$

A.1.3 Stochastic discount factor (SDF) and asset pricing

Define the SDF

$$m_{t,t+1} \equiv \beta \frac{u'(C_{t+1})}{u'(C_t)}. \quad (90)$$

Dividing (87) by $u'(C_t)$ yields

$$1 = \mathbb{E}_t [m_{t,t+1} R_{t+1}^f], \quad (91)$$

and dividing (89) by $u'(C_t)$ yields

$$P_t = \mathbb{E}_t [m_{t,t+1} (D_{t+1} + P_{t+1})]. \quad (92)$$

A.1.4 Transversality conditions

A sufficient transversality condition for optimality is:

$$\lim_{T \rightarrow \infty} \mathbb{E}_0 [\beta^T u'(C_T) A_{T+1}] = 0, \quad \lim_{T \rightarrow \infty} \mathbb{E}_0 [\beta^T u'(C_T) P_T Q_{T+1}] = 0, \quad (93)$$

together with feasibility and any borrowing constraints.

A.2 CRRA Marginal Utility and Euler-Implied Interest Rate

For CRRA utility $u(C) = C^{1-\sigma}/(1-\sigma)$, the marginal utility is $u'(C) = C^{-\sigma}$. Under perfect foresight, (87) implies:

$$C_t^{-\sigma} = \beta C_{t+1}^{-\sigma} R_{t+1}^f \quad \Rightarrow \quad R_{t+1}^f = \frac{1}{\beta} \left(\frac{C_{t+1}}{C_t} \right)^\sigma. \quad (94)$$

Thus,

$$r_{t+1} = R_{t+1}^f - 1 + \delta = \frac{1}{\beta} \left(\frac{C_{t+1}}{C_t} \right)^\sigma - 1 + \delta. \quad (95)$$

A.3 Firm Static Problem: Labor Demand and Envelope Identities

Fix (s, τ, k) and prices (w_t, r_t) . The firm chooses n to maximize

$$\max_{n \geq 0} \left\{ (1 - \tau) s k^\alpha n^\gamma - w_t n \right\}. \quad (96)$$

FOC:

$$(1 - \tau) s k^\alpha \gamma n^{\gamma-1} = w_t. \quad (97)$$

Solving:

$$n_t(s, \tau, k) = \left(\frac{\gamma(1 - \tau)s}{w_t} \right)^{\frac{1}{1-\gamma}} k^{\frac{\alpha}{1-\gamma}}. \quad (98)$$

Then output is

$$y_t(s, \tau, k) = s k^\alpha n_t(s, \tau, k)^\gamma. \quad (99)$$

The envelope identity for Cobb-Douglas implies

$$w_t n_t(s, \tau, k) = \gamma(1 - \tau) y_t(s, \tau, k), \quad (100)$$

so operating profit (revenue net of labor) is

$$(1 - \tau) y_t - w_t n_t = (1 - \tau) y_t - \gamma(1 - \tau) y_t = (1 - \gamma)(1 - \tau) y_t. \quad (101)$$

Hence static profit (net of capital rental and fixed cost) is

$$\pi_t(s, \tau, k) = (1 - \gamma)(1 - \tau) y_t(s, \tau, k) - r_t k - c_f. \quad (102)$$

A.4 Dynamic Firm Problem: Bellman, Timing, and Euler-Style Condition

A.4.1 Bellman equation with household SDF

Let $i_t = k' - (1 - \delta)k$ and adjustment cost $\Phi(i_t, k) \geq 0$. The firm's Bellman equation is

$$V_t(s, \tau, k) = \max_{k' \geq 0} \left\{ \pi_t(s, \tau, k) - i_t - \Phi(i_t, k) + (1 - \lambda) \mathbb{E}_t [m_{t,t+1} V_{t+1}(s, \tau, k')] \right\}. \quad (103)$$

Define the objective as

$$\mathcal{J}_t(k'; s, \tau, k) \equiv \pi_t(s, \tau, k) - i_t - \Phi(i_t, k) + (1 - \lambda)\mathbb{E}_t[m_{t,t+1}V_{t+1}(s, \tau, k')]. \quad (104)$$

A.4.2 FOC (if interior) and marginal value of installed capital

Assume differentiability and an interior solution for k' . Let Φ_i denote the derivative of Φ with respect to i . Since $i_t = k' - (1 - \delta)k$, $\partial i_t / \partial k' = 1$. The FOC is:

$$-1 - \Phi_i(i_t, k) + (1 - \lambda)\mathbb{E}_t[m_{t,t+1}V_{t+1,k}(s, \tau, k')] = 0. \quad (105)$$

This can be rearranged as

$$(1 - \lambda)\mathbb{E}_t[m_{t,t+1}V_{t+1,k}(s, \tau, k')] = 1 + \Phi_i(i_t, k). \quad (106)$$

This condition is the firm analogue of an Euler equation: marginal cost of installing one more unit of k' equals the SDF-discounted marginal continuation value.

Note on discrete grids. In numerical work with a discrete k grid, the policy is typically computed by direct maximization over k' choices rather than imposing (105). The Bellman form (103) remains the correct reference.

A.5 Entry Condition: SDF Form and Complementarity

Entrants at time t pay real cost c_e per unit of entry and appear at $t + 1$ with capital k_0 and type $(s, \tau) \sim g_t$. Because incumbent firms die exogenously with probability λ , the probability the entrant survives into $t + 1$ is $(1 - \lambda)$. The present value to the household of one entrant is therefore

$$\text{PV}_t^{\text{entry}} = (1 - \lambda)\mathbb{E}_t[m_{t,t+1} \mathbb{E}_{(s,\tau) \sim g_t}[V_{t+1}(s, \tau, k_0)]]]. \quad (107)$$

Free entry with complementarity is

$$E_t \geq 0, \quad c_e \geq \text{PV}_t^{\text{entry}}, \quad E_t \cdot (c_e - \text{PV}_t^{\text{entry}}) = 0. \quad (108)$$

If $E_t > 0$, then

$$c_e = (1 - \lambda)\mathbb{E}_t[m_{t,t+1} \mathbb{E}_{g_t} V_{t+1}(s, \tau, k_0)]. \quad (109)$$

A.6 Law of Motion for the Firm Distribution

Let μ_t denote the measure over (s, τ, k) at the start of period t . Given the policy $k'_t(s, \tau, k)$ define the transition map $T_t(s, \tau, k) = (s, \tau, k'_t(s, \tau, k))$.

For any measurable set $A \subset S \times T \times \mathbb{R}_+$,

$$\mu_{t+1}(A) = (1 - \lambda) \int \mathbb{1}\{T_t(x) \in A\} \mu_t(dx) + E_t \int \mathbb{1}\{(s, \tau, k_0) \in A\} g_t(s, \tau) ds d\tau. \quad (110)$$

Total firm mass is $M_t \equiv \mu_t(S \times T \times \mathbb{R}_+)$. Integrating (110) over all states yields the mass recursion

$$M_{t+1} = (1 - \lambda)M_t + E_t. \quad (111)$$

A.7 Aggregation: Detailed Definitions and Consistency Checks

Given (w_t, r_t) and μ_t , define:

$$N_t \equiv \int n_t(s, \tau, k) \mu_t(ds d\tau dk), \quad (112)$$

$$K_t \equiv \int k \mu_t(ds d\tau dk), \quad (113)$$

$$Y_t \equiv \int (sk^\alpha n_t(s, \tau, k)^\gamma - c_f) \mu_t(ds d\tau dk), \quad (114)$$

$$I_t^{\text{inc}} \equiv \int i_t(s, \tau, k) \mu_t(ds d\tau dk), \quad (115)$$

$$\Phi_t^{\text{agg}} \equiv \int \Phi(i_t(s, \tau, k), k) \mu_t(ds d\tau dk). \quad (116)$$

Labor market clearing is $N_t = 1$.

Goods feasibility is

$$C_t + I_t^{\text{inc}} + \Phi_t^{\text{agg}} + c_e E_t = Y_t. \quad (117)$$

A.8 Dividends, Firm-Sector Accounting, and Equity Pricing

A.8.1 Dividends as residual cash flow

A convenient definition of aggregate dividends is:

$$D_t \equiv \int \left(\pi_t(s, \tau, k) - i_t(s, \tau, k) - \Phi(i_t(s, \tau, k), k) \right) \mu_t(ds d\tau dk) - c_e E_t. \quad (118)$$

This definition treats investment, adjustment costs, and entry costs as internal uses of resources prior to payout.

A.8.2 Equity price recursion

The household equity pricing condition implies

$$P_t = \mathbb{E}_t [m_{t,t+1}(D_{t+1} + P_{t+1})]. \quad (119)$$

Under perfect foresight,

$$P_t = m_{t,t+1}(D_{t+1} + P_{t+1}). \quad (120)$$

Given a terminal condition (e.g., convergence to a post-reform steady state), (120) can be solved backward along the transition.

A.8.3 Equity return and pricing identity

Define gross equity return

$$R_{t+1}^e \equiv \frac{D_{t+1} + P_{t+1}}{P_t}. \quad (121)$$

Then (119) is equivalent to:

$$1 = \mathbb{E}_t [m_{t,t+1} R_{t+1}^e]. \quad (122)$$

A.9 Government Closure and Transfers

Wedge revenue is

$$R_t^\tau \equiv \int \tau s k^\alpha n_t(s, \tau, k)^\gamma \mu_t(ds d\tau dk). \quad (123)$$

A balanced-budget lump-sum transfer closure is:

$$T_t = R_t^\tau. \quad (124)$$

Because labor supply is inelastic, this transfer does not distort allocations.

A.10 Steady State Characterizations

In a stationary equilibrium with constant entrant distribution g and constant wedges, all aggregate objects are time-invariant:

$$(C, w, r, P, D, E, \mu).$$

The risk-free Euler equation (91) implies

$$R^f = \frac{1}{\beta} \Rightarrow r = \frac{1}{\beta} - 1 + \delta, \quad (125)$$

for CRRA utility under certainty.

The equity pricing recursion becomes

$$P = m(D + P) = \beta(D + P), \quad (126)$$

so

$$P = \frac{\beta}{1 - \beta} D. \quad (127)$$

Entry with complementarity satisfies

$$c_e \geq (1 - \lambda)\beta \mathbb{E}_g[V(s, \tau, k_0)], \quad E \cdot \left(c_e - (1 - \lambda)\beta \mathbb{E}_g V(s, \tau, k_0) \right) = 0. \quad (128)$$

A.11 Residual Definitions for Numerical Verification

For numerical solutions, it is useful to compute residuals that directly correspond to equilibrium conditions.

Goods feasibility residual.

$$\mathcal{R}_t^{\text{goods}} \equiv (C_t + I_t^{\text{inc}} + \Phi_t^{\text{agg}} + c_e E_t) - Y_t. \quad (129)$$

Labor market residual.

$$\mathcal{R}_t^{\text{labor}} \equiv N_t - 1. \quad (130)$$

Risk-free Euler residual.

$$\mathcal{R}_t^{\text{Euler},f} \equiv u'(C_t) - \beta u'(C_{t+1}) R_{t+1}^f. \quad (131)$$

Equity Euler residual.

$$\mathcal{R}_t^{\text{Euler},e} \equiv u'(C_t) P_t - \beta u'(C_{t+1}) (D_{t+1} + P_{t+1}). \quad (132)$$

Entry complementarity residual. Define the free-entry gap:

$$\mathcal{G}_t^{\text{entry}} \equiv (1 - \lambda) \mathbb{E}_t \left[m_{t,t+1} \mathbb{E}_{g_t} V_{t+1}(s, \tau, k_0) \right] - c_e. \quad (133)$$

Then complementarity requires $E_t \geq 0$, $\mathcal{G}_t^{\text{entry}} \leq 0$, and

$$E_t \cdot \mathcal{G}_t^{\text{entry}} = 0. \quad (134)$$

Distribution consistency check. Given μ_{t+1}^{sim} produced by applying (110) to (μ_t, k'_t, E_t) , check $\|\mu_{t+1} - \mu_{t+1}^{\text{sim}}\|$ is numerically small.

B Replication and Extension Code and Output

This appendix contains the full Jupyter Notebooks used to replicate the original model and implement the transition dynamics extension. All relevant output is also included. Note that the first notebook contains the replication and an illustration of

<https://github.com/silentsoundss/RR-Transition-Dynamics/tree/main>

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